

Auto Grader Report

UAID: 300349320

Course: CSCE 2004

Task: lab7c

Syntax - Correct

Program source:

```
void print_roots(double a, double b, double c) {
    double discriminant = b * b - 4 * a * c;
    double root1 = (-b - sqrt(discriminant)) / (2 * a);
    double root2 = (-b + sqrt(discriminant)) / (2 * a);

    if (discriminant < 0)
        cout << "There are no roots to this quadratic equation\n";
    else if (discriminant == 0)
        cout << "There is one root: " << root1 << endl;
    else
        cout << "There are two roots: " << root1 << " and " << root2 << endl;
}

int main() {
    double a = 0, b = 0, c = 0;

    // Get coefficients
    get_coefficients(a, b, c);

    // Print original equation and roots
    print_quadratic(a, b, c);
    print_roots(a, b, c);

    // Print negated equation and roots
    print_quadratic(-a, -b, -c);
    print_roots(-a, -b, -c);

    return 0;
}
```

Test #1 - Correct

Test input:

1 -5 4

Your program output:

```
Enter quadratic coefficients a b c:
Quadratic equation: 1x^2 + -5x + 4
There are two roots: 1 and 4
Quadratic equation: -1x^2 + 5x + -4
There are two roots: 4 and 1
```

Expected program output:

```
Enter quadratic coefficients a b c:  
Quadratic equation: 1x^2 + -5x + 4  
There are two roots: 1 and 4  
Quadratic equation: -1x^2 + 5x + -4  
There are two roots: 4 and 1
```

Test #2 - Correct**Test input:**

```
1 4 1
```

Your program output:

```
Enter quadratic coefficients a b c:  
Quadratic equation: 1x^2 + 4x + 1  
There are two roots: -3.73205 and -0.267949  
Quadratic equation: -1x^2 + -4x + -1  
There are two roots: -0.267949 and -3.73205
```

Expected program output:

```
Enter quadratic coefficients a b c:  
Quadratic equation: 1x^2 + 4x + 1  
There are two roots: -3.73205 and -0.267949  
Quadratic equation: -1x^2 + -4x + -1  
There are two roots: -0.267949 and -3.73205
```

Score = 100%

Grade Summary

UAID	300349320
lab1a	100%
lab2a	100%
lab2b	100%
lab2c	100%
lab3a	100%
lab3b	100%
lab3c	100%
lab4a	100%
lab4b	100%
lab4c	100%
lab5a	100%
lab5b	100%

lab5c	100%
lab6a	100%
lab6b	100%
lab6c	100%
lab7a	100%
lab7b	100%
lab7c	100%